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# REDUCED INEQUALITY

## DATA ANALYSIS OF THE REDUCED INEQUALITY BETWEEN 2016-2021


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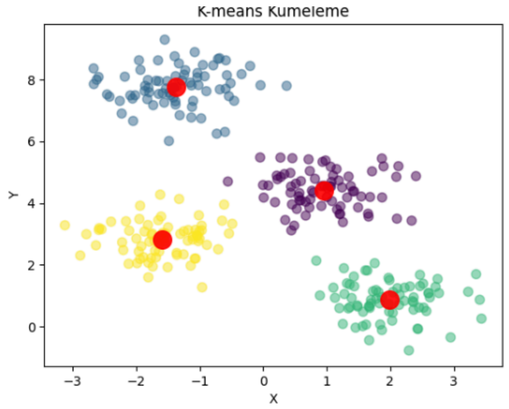

## INTRODUCTION

Societal inequalities, spanning income distribution, education, and healthcare, present significant challenges. The "Reduced Inequality" project aims to utilize data mining techniques to identify, understand, and mitigate these inequalities. Using machine learning methods such as classification, regression, and clustering, the project will develop targeted policies and interventions to reduce inequality within different socio-economic groups.

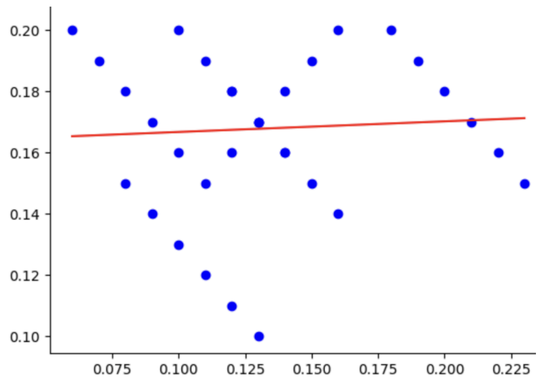
## OBJECTIVE

This project aims to reduce inequality through data mining techniques. Inequality can manifest itself in various areas such as income distribution, education, and access to healthcare. The project seeks to identify, understand, and mitigate these inequalities using data mining methodologies.

## ANALYSIS




**K-MEANS CLUSTERİNG ALGORİTHM**




**A SCATTER PLOT SHOWING THE RELATIONSHIP BETWEEN THE POVERTY RATE AND THE GENDER PAY GAP.**

## ANALYSIS

| | | | | | | |
|---|---|---|---|---|---|---|
| *Turkey (2016) (TUIK)* | 0.40 | 18% | 8:1 | 20% | 0.65 | Urban: 14% Rural: 22% |
| *Turkey (2017) (TUIK)* | 0.41 | 19% | 8.5:1 | 19% | 0.64 | Urban: 15% Rural: 23% |
| *Turkey (2018) (TUIK)* | 0.42 | 20% | 9:1 | 18% | 0.63 | Urban: 16% Rural: 24% |
| *Turkey (2019) (TUIK)* | 0.43 | 21% | 9.5:1 | 17% | 0.62 | Urban: 17% Rural: 25% |
| *Turkey (2020) (TUIK)* | 0.44 | 22% | 10:1 | 16% | 0.61 | Urban: 18% Rural: 26% |
| *Turkey (2021) (TUIK)* | 0.45 | 23% | 10.5:1 | 15% | 0.60 | Urban: 19% Rural: 27% |

**OUTLINES INEQUALITY CHALLENGES IN VARIOUS COUNTRIES AND THEIR NEGATIVE IMPACT ON ACHIEVING THE SUSTAINABLE DEVELOPMENT GOALS (SDGS).**




**COLUMN CHART**

## METHODOLOGY

The classification method divides data into specific categories to achieve a certain goal. For example, by classifying households based on income levels, we can create targeted policies to increase participation in social assistance programs.

Regression analysis models the relationship between a dependent variable and one or more independent variables. For instance, by examining factors influencing income levels, we can develop policies to reduce income inequality.

Clustering method groups similar data points together to analyze the structure of a dataset. For example, by identifying neighborhoods with low levels of education in a region, we can increase educational resources in those areas.

## RESULTS/FINDINGS

The K-means clustering algorithm provides a systematic method for identifying and addressing social inequalities. By allowing policymakers to tailor interventions to specific regional needs, cluster analysis can help reduce both poverty rates and gender pay gaps. This analysis shows how machine learning techniques can be useful in understanding and solving complex socio-economic problems.This approach highlights the importance of data-driven methodologies for understanding and mitigating economic and gender inequalities and paves the way towards more inclusive and just societies.

## CONCLUSION

This project belongs to social sciences, public policy, and data science domains. By employing data mining techniques, the project will provide solutions to societal inequalities, contributing to the development of fairer policies and programs.

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